Use Case 1 & 4

Lambda Function Console:
Functions | Lambda

Lambda Function for Correct Data Scenario -> ProducerDemoApp

Lambda Function for Missing Data Scenario -> Failure-Simulation

CloudWatch Alarms:
Alarms | CloudWatch | us-east-1

Fargate Consumer Logs in Cloudwatch:

FargateConsumer Cloudwatch Logs

EC2 Instance: -

EC2 Instance Connect Link:

PostgresRDS-EC2 Instance Connect Link

Then Connect using a Public IP and click Connect.

Commands to connect to RDS demoapp-telemetry-db **:**

Once you are connected to the same EC2 instance via Connect Link, execute the commands formatted in bold to connect to RDS within the EC2 instance from the location '**/home/ubuntu**'. Enter the password "**Msys#123**" and you will be logged into RDS.

```
ubuntu@ip-10-0-2-158:~$ pwd
/home/ubuntu

ubuntu@ip-10-0-2-158:~$ curl -o global-bundle.pem
https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem

ubuntu@ip-10-0-2-158:~$ export RDSHOST="demoapp-telemetry-db.c1n1vvf1lfpz.us-east-
1.rds.amazonaws.com"

ubuntu@ip-10-0-2-158:~$ psql "host=$RDSHOST port=5432 dbname=postgres
user=demoadmin sslmode=verify-full sslrootcert=./global-bundle.pem"

Password for user demoadmin:
psql (16.13 (Ubuntu 16.13-0ubuntu0.24.04.1), server 17.6)
WARNING: psql major version 16, server major version 17.
        Some psql features might not work.
```

```
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression: off)
Type "help" for help.

postgres=>
```

Once you are inside the Postgres db, execute the below command highlighted in bold so that you can verify the existence of database **'democonsapp'** and then connect to it. Finally, list the records from the **'telemetry'** table.

```
postgres=> \c democonsapp;

You are now connected to database "democonsapp" as user "demoadmin".
democonsapp=>
democonsapp=> select * from telemetry;
```

Step 1: Execute Producer lambda function to generate data via aws cli.

```
aws lambda invoke \
 --function-name ProducerDemoApp \
 --payload '{}' \
 --region us-east-1 \
 /tmp/lambda-output.json
```

The data will be transferred to the Standard Queue **DemoAppTelemetryQueue** and then the generated data will be processed immediately. You will find these kinds of logs 5 times with different MessageIDs in the FargateConsumer Cloudwatch Logs

2026-04-14T23:36:55.417+05:30	INFO:__main__:✓ Message processed successfully: ba0980e0-d6a4-4c

Since the processing of payload is happening within microseconds, so you may not find payload in **DemoAppTelemetryQueue**. That's why you can fetch that payload in **telemetry** table within RDS.

```
democonsapp=> SELECT * FROM telemetry ORDER BY id DESC LIMIT 6;
```

The positive scenario simulation completed successfully.

Step 2: Execute Failure-Simulation lambda Function to generate error data via aws cli.

```
aws lambda invoke \
 --function-name Failure-Simulation \
 --payload '{}' \
 --region us-east-1 \
 /tmp/lambda-output-failure.json
```

The payload will be transferred to the **DemoAppTelemetryDLQ** after FargateConsumer fails to process. You will find logs of 4 kinds in the FargateConsumer Cloudwatch Logs

```
2026-04-14T23:50:26.555+05:30    ERROR:__main__:✗ Validation error for MessageId 2ac36f19-8597-4db0-9534-ce45be84e531: Expecting property name enclosed in double quotes: line 1 column 88 (char 87)
```

Total 4 messages will be lying in the DLQ, and this can be observed through this AWS SQS Link.

```
aws cloudwatch set-alarm-state \
 --alarm-name ProducerErrorAlert \
 --state-value ALARM \
 --state-reason "Testing producer error - Manual trigger" \
 --region us-east-1
```

Step 3: SQS-DLQ will send logs to CloudWatch, CloudWatch will create alert to Jira. Jira will forward the summary of latest tickets in past 1 hour to Aziron/Webhook.

Project Backlog:
openshift - Backlog - Jira

Step 4: Resolution identify that data is missing in SQS-DLQ and update in Jira ticket, create a backup of msg and delete it from SQS-DLQ.

```
aws sqs get-queue-url --queue-name DemoAppTelemetryDLQ --region us-east-1

aws sqs purge-queue --queue-url https://sqs.us-east-1.amazonaws.com/267767410086/DemoAppTelemetryDLQ --region us-east-1
```

Note: The error message ID is provided in the Jira ticket, which AZIRON can use to investigate and resolve the issue. The error will be of missing values in data received from the producer.